SOIL-STRUCTURE INTERACTION

1. Introduction to Nonlinear Behaviour of Soil Materials

   Simplified approach of soil response with linear and nonlinear springs, fundamental assumptions, limits and fields of application. Failure Criteria, yielding, flow rules, elastoplasticity, constitutive equations of perfect plasticity and plasticity with hardening. Element stiffness matrix, assembly of global stiffness matrix, global force vector. From linear or linear elasticity to a system of linear equations.

2. Principles of Simulating Structures

   Simulation of actions (gravity loads, seismic action, initial stress condition). Discretisation, principles of simulating constitutive elements of a structure. Simplified and complex constitutive models, (concrete, steel, adhesion between steel-concrete). Response of cracked element under monotonic, repeated and reverse loading.

3. Simulation of Soil-Structure using Numerical Methods

   Methodology of problem approach. Principles of discretisation, and simulation of representative soil-structure cases. Geometrical and loading assumptions in approaching 3D problems with plane strain or axisymmetric conditions. Multistage simulation process of problem with variable domain and boundaries.

4. Applications

   Retaining structures in urban areas and highway projects. Particular foundations of buildings and infrastructures. Tunnels and underground structures. Response of Dams under construction stages. The use of monitoring and measurements in the design of civil engineering projects.